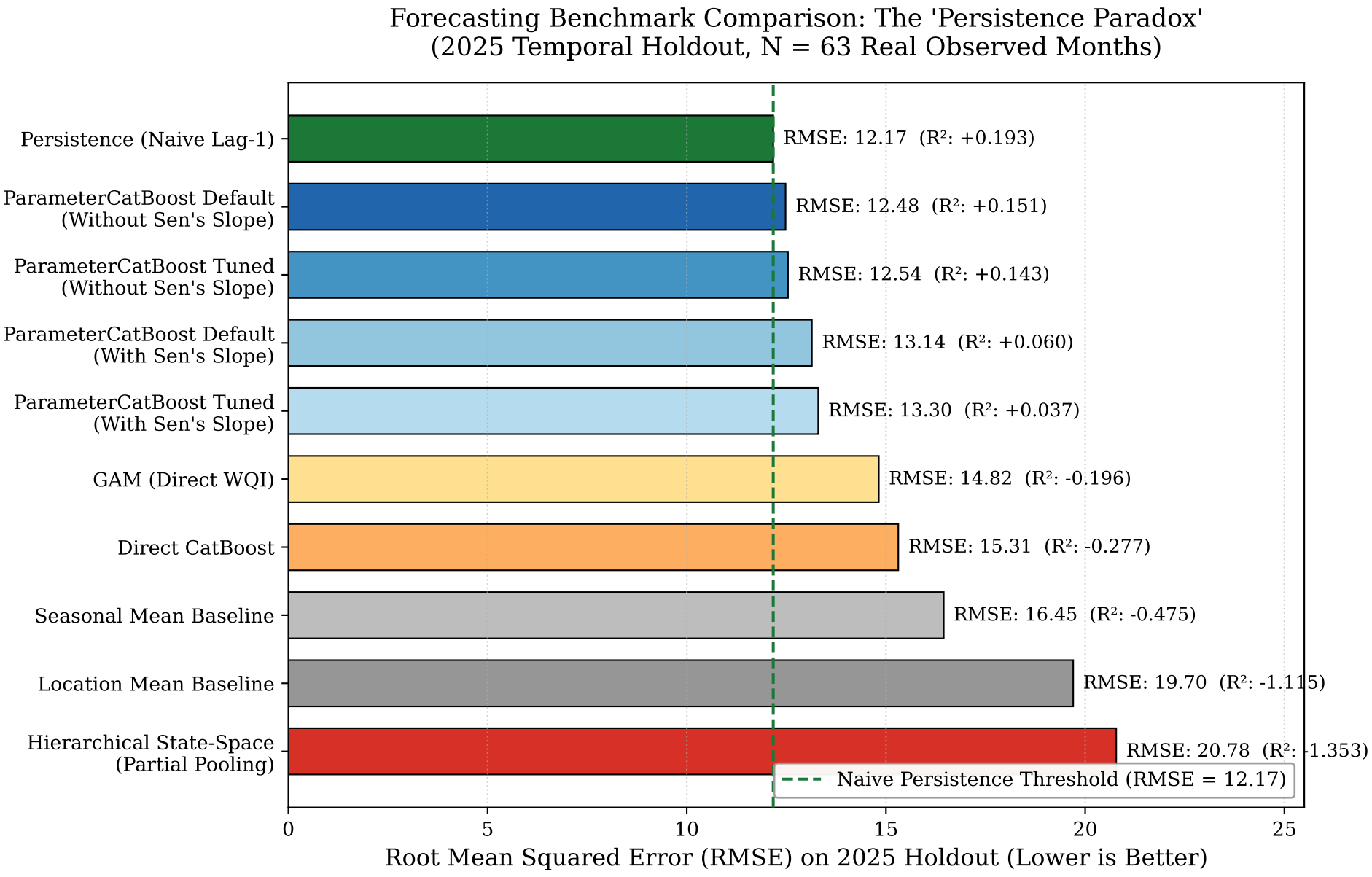
Forecasting Benchmark Comparison: The 'Persistence Paradox'
(2025 Temporal Holdout, N = 63 Real Observed Months)
Persistence (Naive Lag-1)
RMSE: 12.17 (R²: +0.193)
ParameterCatBoost Default
(Without Sen's Slope)
RMSE: 12.48 (R²: +0.151)
ParameterCatBoost Tuned
(Without Sen's Slope)
RMSE: 12.54 (R²: +0.143)
ParameterCatBoost Default
(With Sen's Slope)
RMSE: 13.14 (R²: +0.060)
ParameterCatBoost Tuned
(With Sen's Slope)
RMSE: 13.30 (R²: +0.037)
GAM (Direct WQI)
RMSE: 14.82 (R²: -0.196)
Direct CatBoost
RMSE: 15.31 (R²: -0.277)
Seasonal Mean Baseline
RMSE: 16.45 (R²: -0.475)
Location Mean Baseline
RMSE: 19.70 (R²: -1.115)
Hierarchical State-Space
(Partial Pooling)
RMSE: 20.78 (R²: -1.353)
Naive Persistence Threshold (RMSE = 12.17)
0
5
10
15
20
25
Root Mean Squared Error (RMSE) on 2025 Holdout (Lower is Better)